Xylem Pipeline Relay

Move water uphill like a tree, faster and cleaner than rival teams.

Win condition: Most water delivered with speed, low spill, and strong explanation.

THE SCIENCE YOU ARE USING

Capillary action. Water climbs narrow spaces on its own. Water molecules stick to a surface (adhesion) and to each other (cohesion), so a thin column is pulled upward against gravity. Narrower channels and more wettable materials pull water higher and faster. Trees use this in their narrow xylem, but on its own it lifts water only about a meter. To reach the treetop, water evaporating from the leaves (transpiration) pulls the whole column up, like sipping through a straw.

Material and geometry. A folded paper towel, a cotton ball pulled into a strand, and a strip of cotton cloth each move water at a different rate. Capillary action only works in fine, connected channels that water can wet, so pre-wet your wick, keep it in firm contact, and use a gentle, short climb. Path length, slope, and contact all matter. Your team picks the material and route to deliver the most water to a target cup.

HOW TO BUILD AND RUN IT

01 **Test three wick materials.** Pre-wet each wick, then dip the paper towel, the cotton-ball strand, and the cotton cloth into colored water. Watch which climbs fastest and highest.

02 **Pick a material and route.** Choose your wick and lay out a path from the source cup up over a block to the target cup.

03 **Run the first transfer.** Start the wick and time how much water reaches the target in the time limit. Catch spills on the bag-covered table.

04 **Reduce spill and length.** Trim the path, improve contact, and re-run. Less spill and a shorter path usually deliver more.

05 **Final delivery round.** Run the best design for score. Water delivered, speed, and low spill all count.

Safety: Work over the bag-covered table and wipe spills immediately so the floor stays dry. No standing on chairs or stacked furniture to build height. Use washable food coloring; it can stain skin and clothes, so wear gloves if you do not want stained hands. No goggles are needed for moving water.

MATERIALS

Paper towels	shared (about 1 roll)
Cotton balls	a few per team (pull into strands)
Cotton cloth strips	3 per team (16 total)
Cups	2 per team (8)
Food coloring	shared (washable)
Table cover (trash bag)	1 per team (4)
Elevation blocks	1 per team (4)
Rulers	1 per team (4)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ____________________

Wick we chose (paper towel / cotton ball / cotton cloth) and why: ____________________

The ONE thing we changed for the redesign (path length, slope, contact, or material): ____________________

Water delivered: baseline ________ **after redesign** ________ **spill** ________

DEFEND YOUR DESIGN

What evidence made you change your design? ____________________

If you ran it again, what is the ONE thing you would change? ____________________

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Water delivered	40
Speed	20
Least spill	15
Design explanation	15
Redesign gain	10
Total	**100**